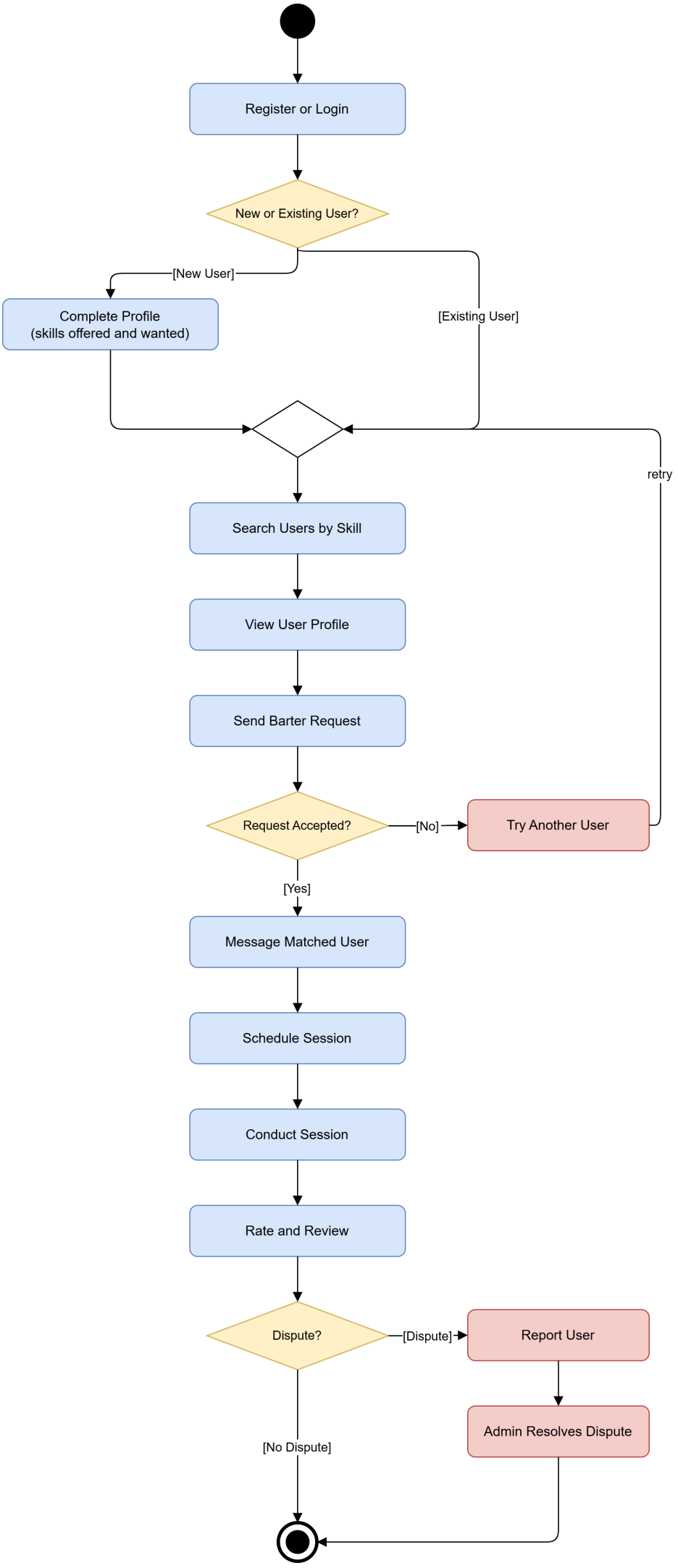
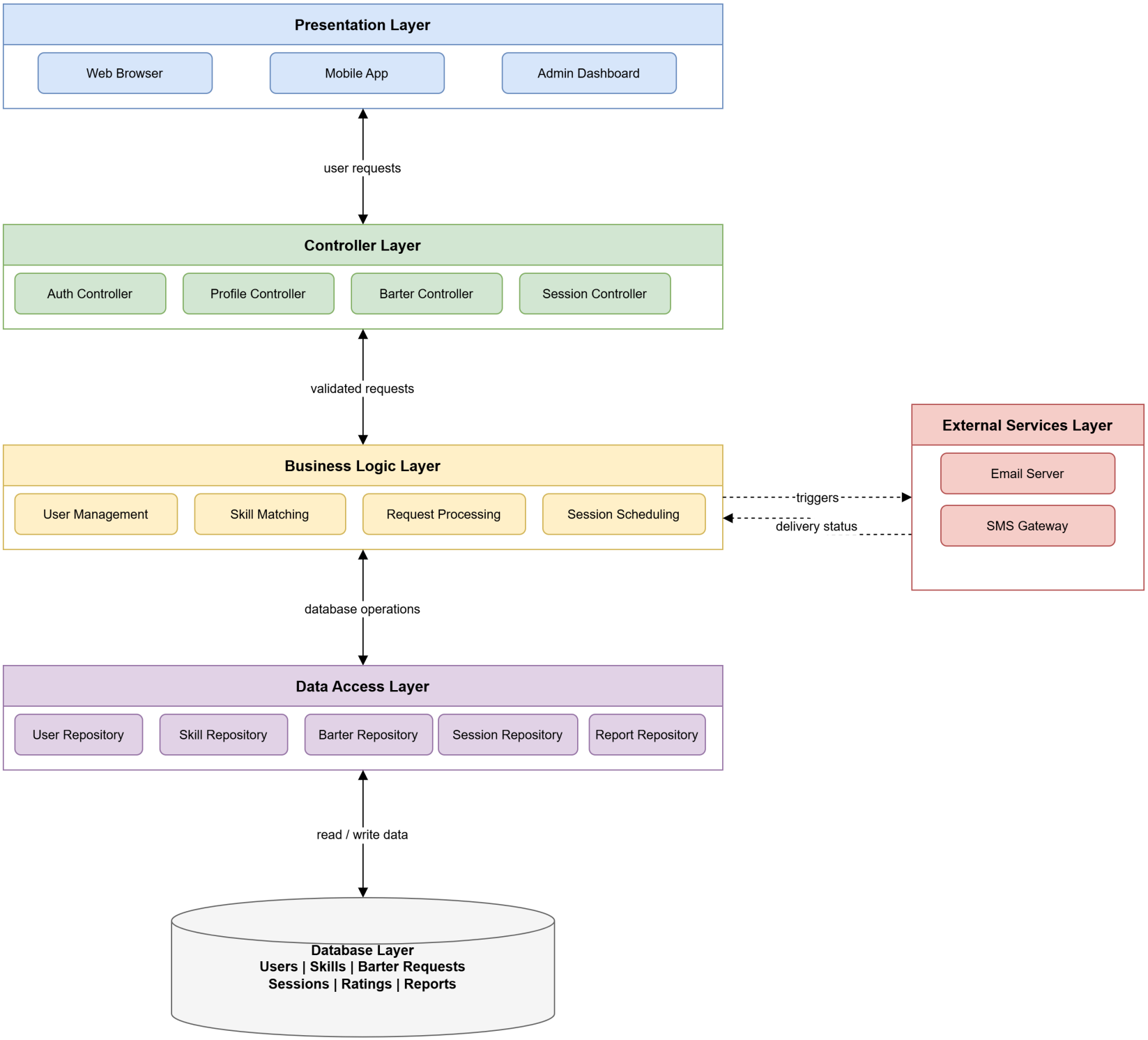
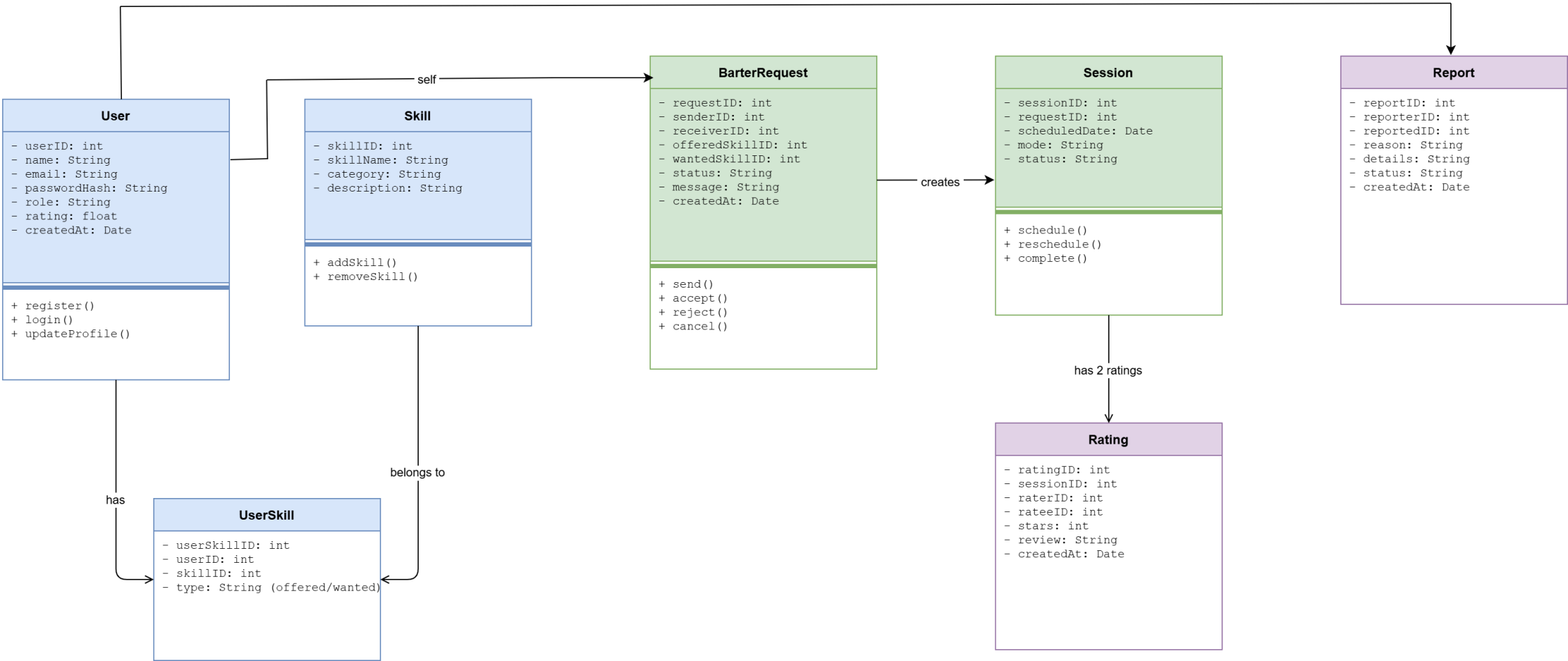

Register or Login
New or Existing User?
[New User]
Complete Profile
(skills offered and wanted)
[Existing User]
retry
Search Users by Skill
View User Profile
Send Barter Request
Request Accepted?
[No]
Try Another User
[Yes]
Message Matched User
Schedule Session
Conduct Session
Rate and Review
Dispute?
[Dispute]
Report User
Admin Resolves Dispute
[No Dispute]


Presentation Layer
Web Browser
Mobile App
Admin Dashboard
user requests
Controller Layer
Auth Controller
Profile Controller
Barter Controller
Session Controller
validated requests
Business Logic Layer
User Management
Skill Matching
Request Processing
Session Scheduling
triggers
delivery status
External Services Layer
Email Server
SMS Gateway
database operations
Data Access Layer
User Repository
Skill Repository
Barter Repository
Session Repository
Report Repository
read / write data
Database Layer
Users | Skills | Barter Requests
Sessions | Ratings | Reports


User
- userID: int
- name: String
- email: String
- passwordHash: String
- role: String
- rating: float
- createdAt: Date
+ register()
+ login()
+ updateProfile()
Skill
- skillID: int
- skillName: String
- category: String
- description: String
+ addSkill()
+ removeSkill()
self
BarterRequest
- requestID: int
- senderID: int
- receiverID: int
- offeredSkillID: int
- wantedSkillID: int
- status: String
- message: String
- createdAt: Date
+ send()
+ accept()
+ reject()
+ cancel()
creates
Session
- sessionID: int
- requestID: int
- scheduledDate: Date
- mode: String
- status: String
+ schedule()
+ reschedule()
+ complete()
Report
- reportID: int
- reporterID: int
- reportedID: int
- reason: String
- details: String
- status: String
- createdAt: Date
has
belongs to
UserSkill
- userSkillID: int
- userID: int
- skillID: int
- type: String (offered/wanted)
has 2 ratings
Rating
- ratingID: int
- sessionID: int
- raterID: int
- rateeID: int
- stars: int
- review: String
- createdAt: Date

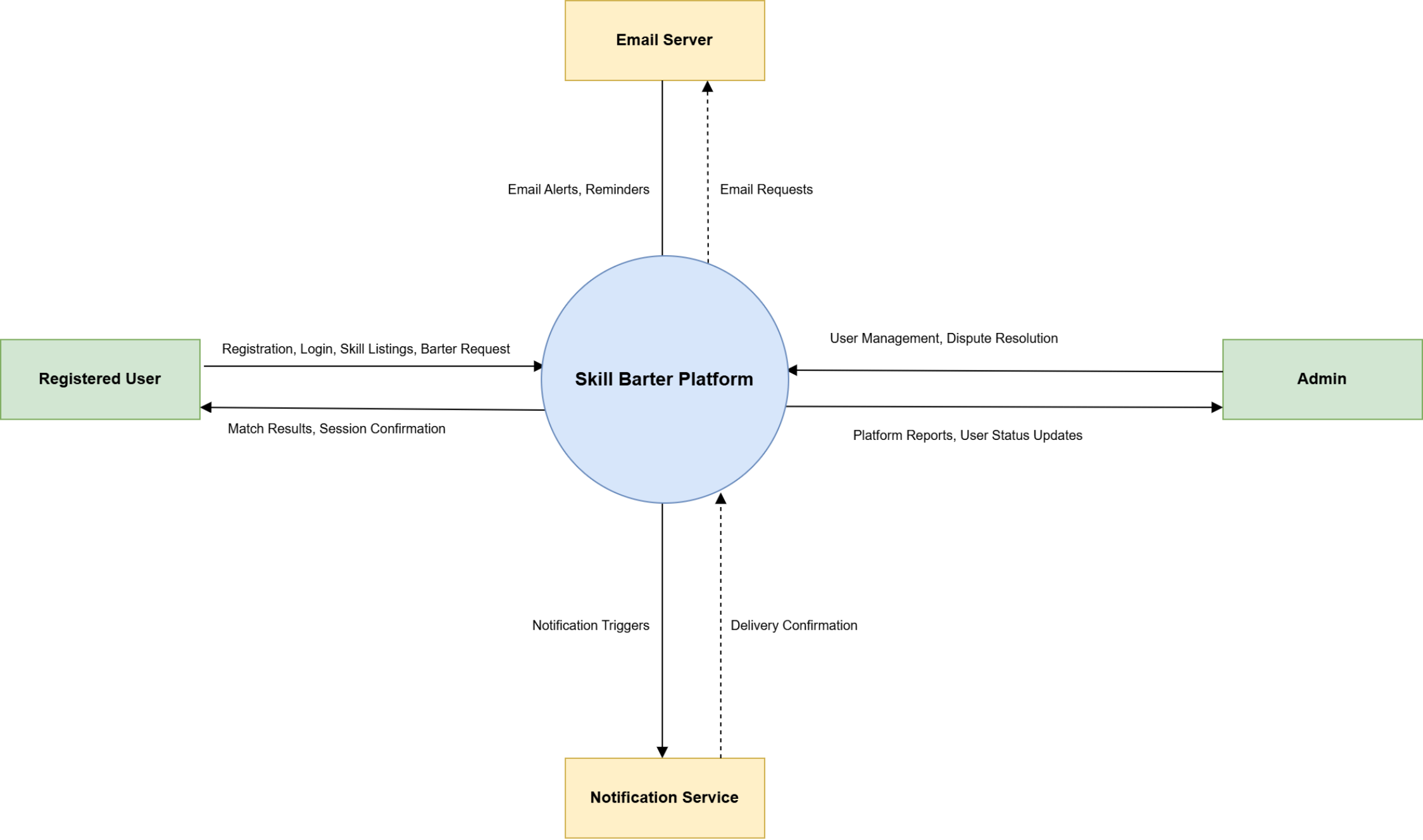
Commits

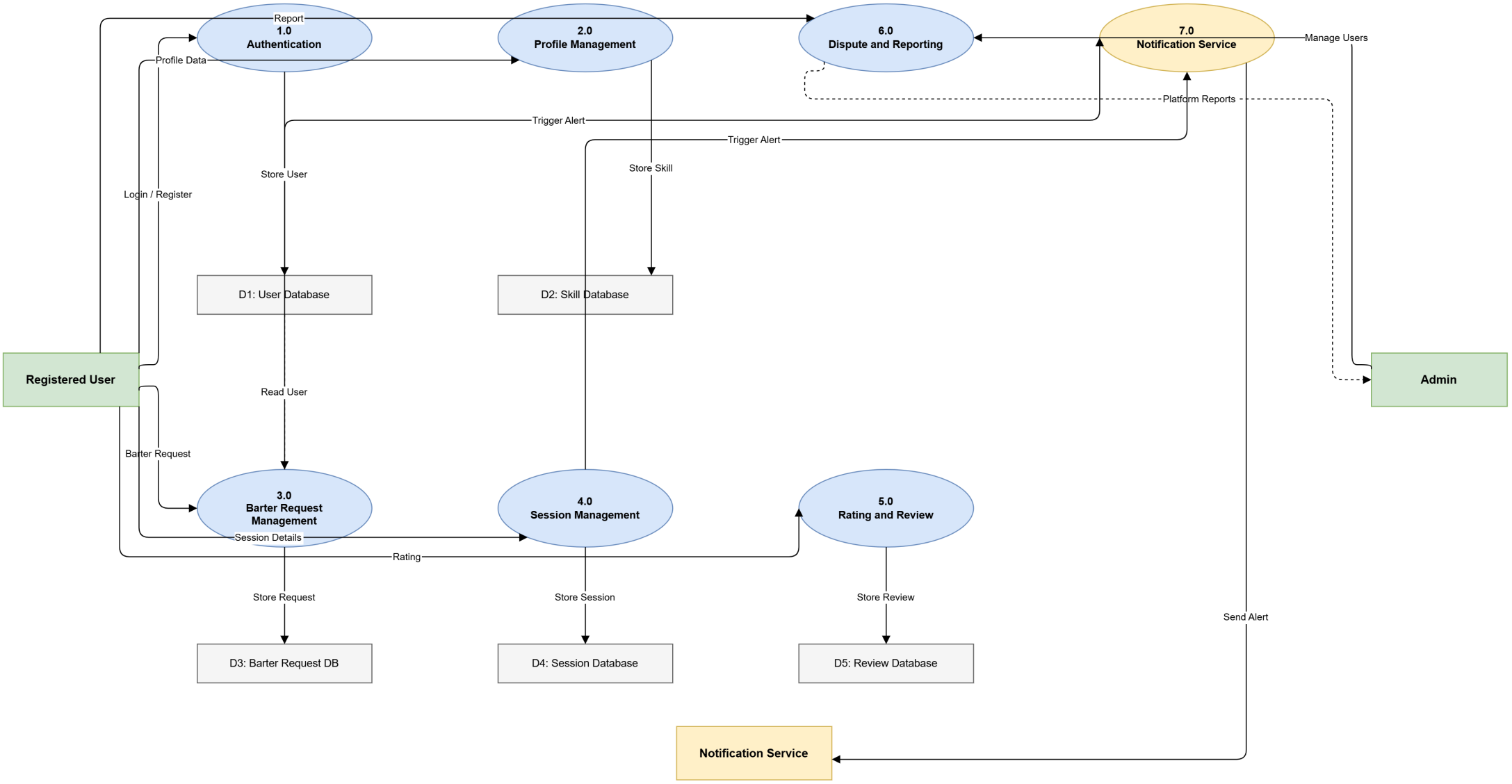
main

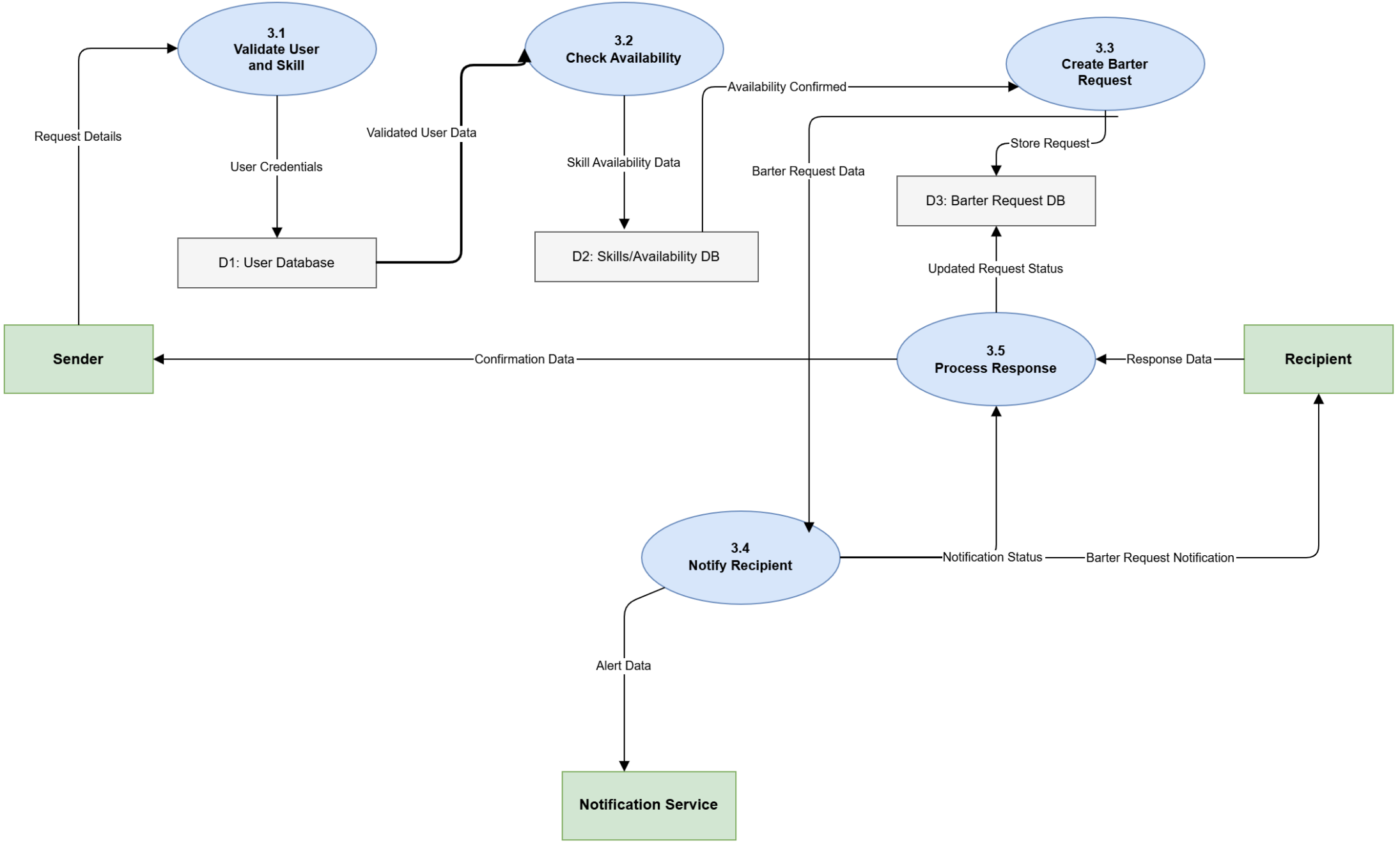
All users | All time

Commits on May 9, 2026

Add files via upload JunaidAhmed13 authored 3 minutes ago	Verified	251ca65
Add files via upload JunaidAhmed13 authored 5 minutes ago	Verified	5cc3eb7
Create README.md JunaidAhmed13 authored 11 minutes ago	Verified	bd7d0cc
Update README.md JunaidAhmed13 authored 14 minutes ago	Verified	fb8f46f
Initial commit JunaidAhmed13 authored 14 minutes ago	Verified	a0d6374


Email Server
Email Alerts, Reminders
Email Requests
Registered User
Registration, Login, Skill Listings, Barter Request
Match Results, Session Confirmation
Skill Barter Platform
User Management, Dispute Resolution
Platform Reports, User Status Updates
Admin
Notification Triggers
Delivery Confirmation
Notification Service


1.0
Authentication
2.0
Profile Management
6.0
Dispute and Reporting
7.0
Notification Service
Report
Profile Data
Manage Users
Platform Reports
Trigger Alert
Trigger Alert
Store User
Store Skill
Login / Register
D1: User Database
D2: Skill Database
Registered User
Admin
Read User
Barter Request
3.0
Barter Request
Management
4.0
Session Management
5.0
Rating and Review
Session Details
Rating
Store Request
Store Session
Store Review
Send Alert
D3: Barter Request DB
D4: Session Database
D5: Review Database
Notification Service


3.1
Validate User
and Skill
3.2
Check Availability
3.3
Create Barter
Request
Request Details
User Credentials
Validated User Data
Skill Availability Data
Availability Confirmed
Barter Request Data
Store Request
D3: Barter Request DB
D1: User Database
D2: Skills/Availability DB
Updated Request Status
Sender
Confirmation Data
3.5
Process Response
Response Data
Recipient
3.4
Notify Recipient
Notification Status
Barter Request Notification
Alert Data
Notification Service

Skill Barter Platform -- Project Gantt Chart (4 Weeks)

Task	Week 1	Week 2	Week 3	Week 4	Owner
M1: Problem Statement					Sadiq
M1: Functional Requirements					Junaid
M1: Use Case Diagrams					Bilawal
M2: Context and DFD					Shah Murad
M2: Class Diagram					Junaid
M2: Behavioral Diagrams					Bilawal
M3: Architecture Design					Sadiq
M3: UML Refinement					All Members
M4: Test Plan					Shah Murad
M4: GitHub and Version Control					Sadiq
M4: Risk Analysis and Planning					Junaid

Milestone 1 Milestone 2 Milestone 3 Milestone 4

JunaidAhmed13 / Skill-Barter-Platform

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security and quality | Insights | Settings

Skill-Barter-Platform Public

Pin | Watch 0 | Fork 0 | Star 0

main | 1 Branch | 0 Tags | Go to file T | Add file | Code

JunaidAhmed13 Add files via upload 251ca65 · 2 minutes ago 5 Commits

diagrams	Add files via upload	4 minutes ago
README.md	Update README.md	13 minutes ago
diagrams.pdf	Add files via upload	2 minutes ago

README

Skill Barter Platform

A software engineering documentation project developed as part of SEN 220 at Bahria University, Islamabad.

Project Overview

The Skill Barter Platform is a web-based system that allows users to exchange skills directly without any monetary transaction. Users list skills they offer and skills they want, find matching users, and arrange skill swap sessions.

Team Members

About

Software Engineering documentation project for a Skill Barter Platform

Readme
Activity
0 stars
0 watching
0 forks

Releases

No releases published
Create a new release

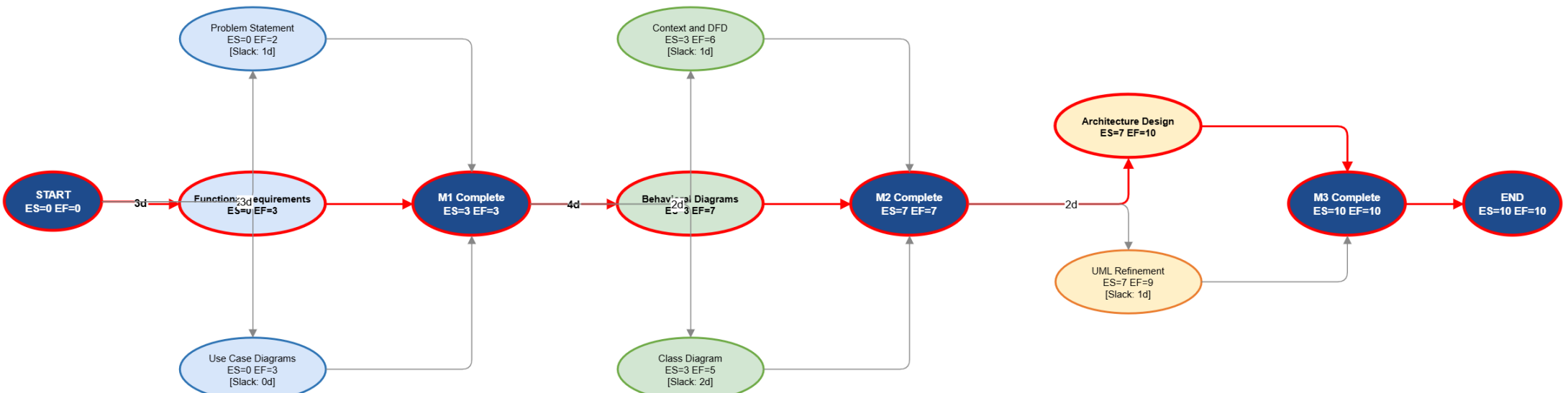
Packages

No packages published
Publish your first package

Contributors 1

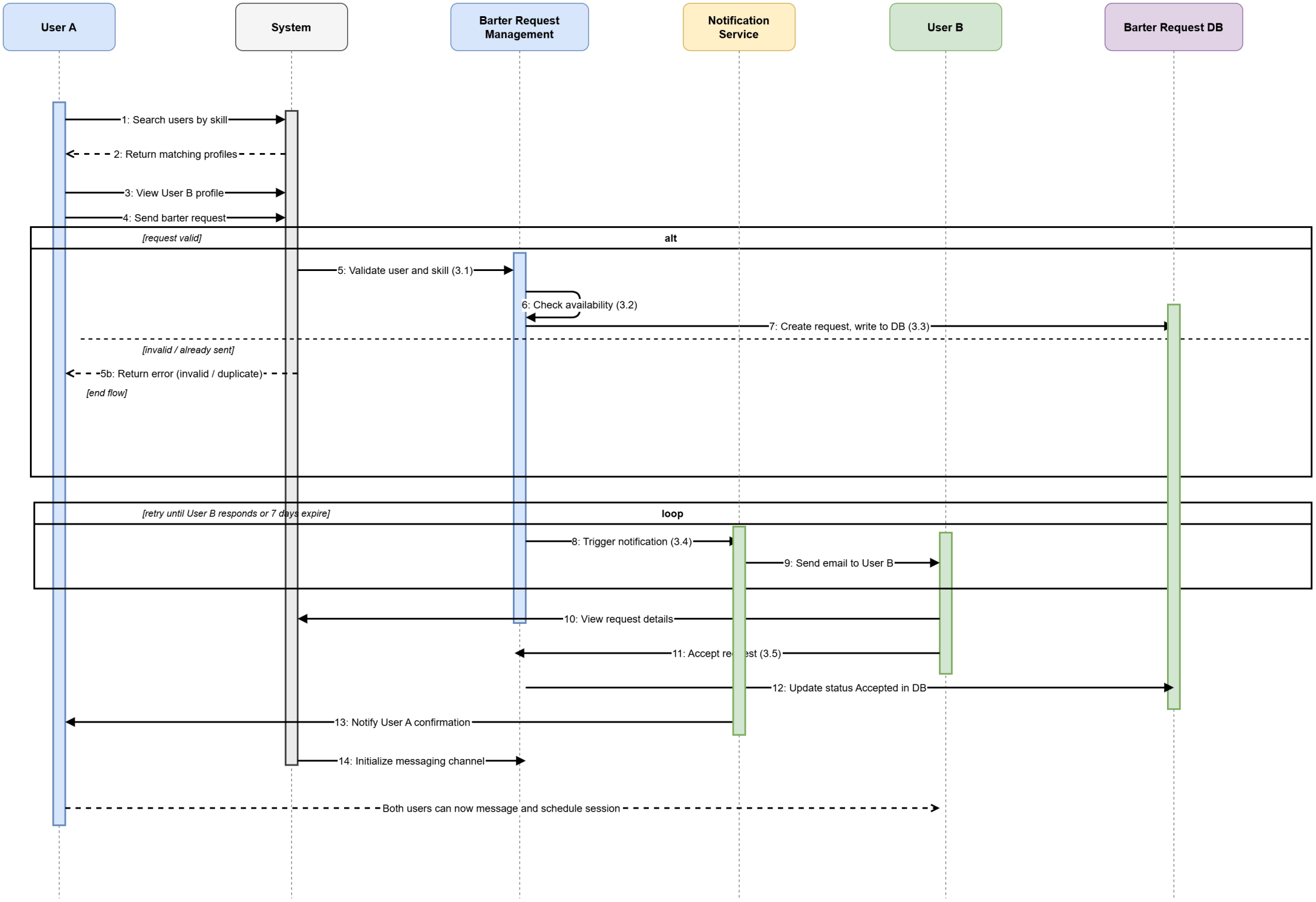
JunaidAhmed13

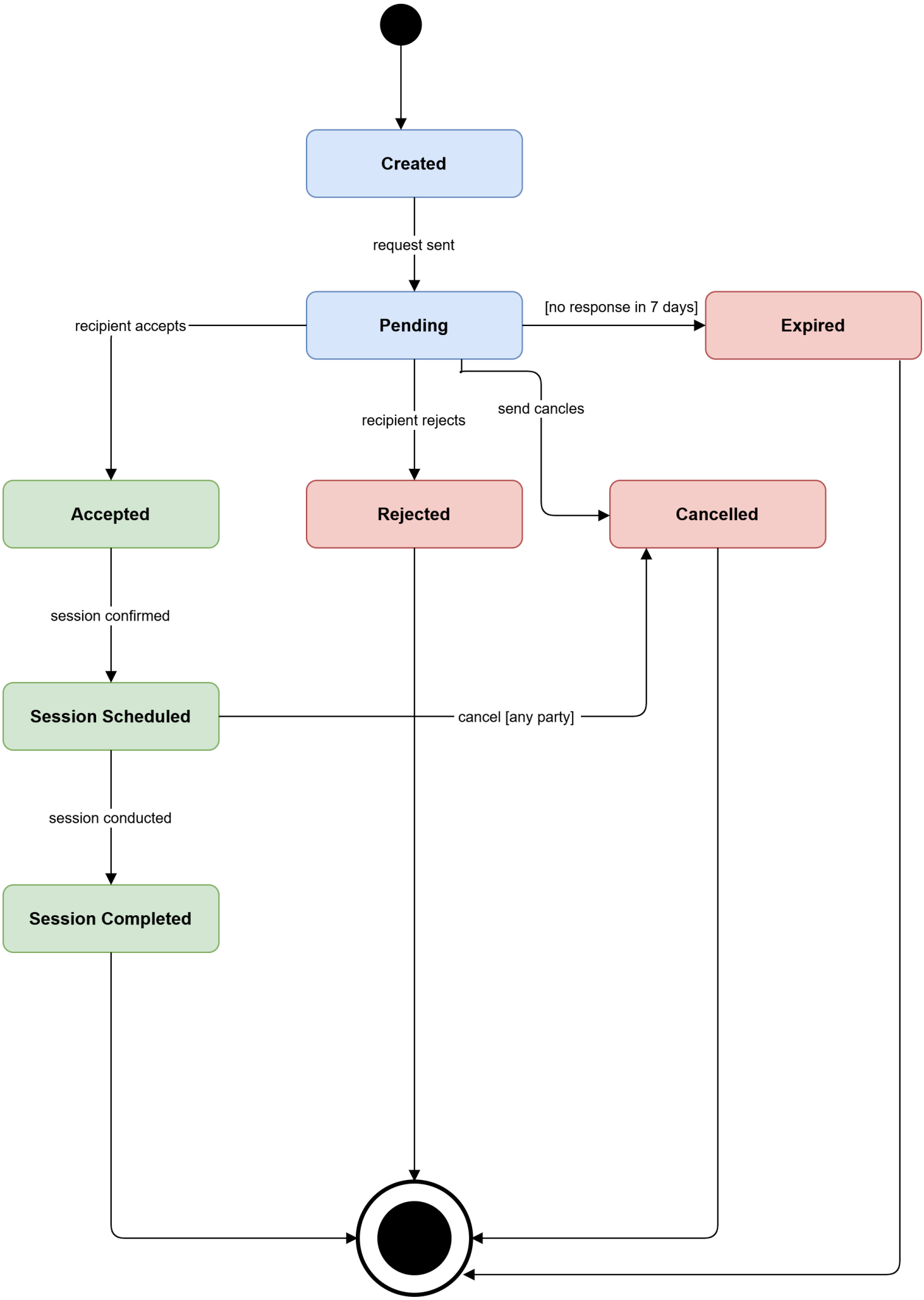
Skill Barter Platform -- PERT Chart (Activity on Node)

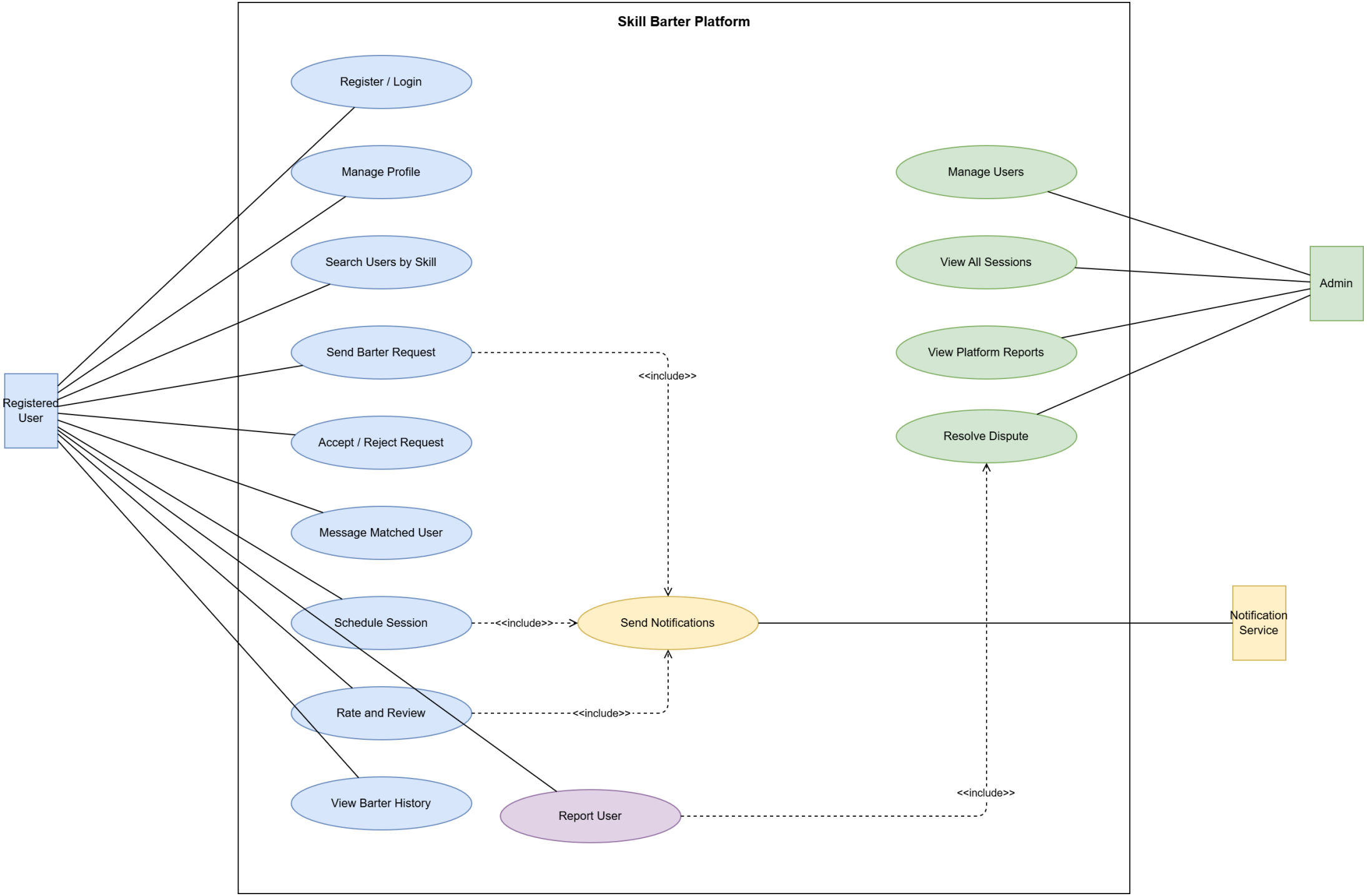


Total Project Duration: 10 working days | Critical Path: Start > Functional Requirements (3d) > M1 Complete > Behavioral Diagrams (4d) > M2 Complete > Architecture Design (3d) > M3 Complete > End

Critical Path | Non-Critical | Milestone 1 | Milestone 2 | Milestone 3


User A
System
Barter Request Management
Notification Service
User B
Barter Request DB
1: Search users by skill
2: Return matching profiles
3: View User B profile
4: Send barter request
[request valid]
alt
5: Validate user and skill (3.1)
6: Check availability (3.2)
7: Create request, write to DB (3.3)
[invalid / already sent]
5b: Return error (invalid / duplicate)
[end flow]
[retry until User B responds or 7 days expire]
loop
8: Trigger notification (3.4)
9: Send email to User B
10: View request details
11: Accept request (3.5)
12: Update status Accepted in DB
13: Notify User A confirmation
14: Initialize messaging channel
Both users can now message and schedule session


Created
request sent
Pending
[no response in 7 days]
Expired
recipient accepts
send cancles
recipient rejects
Accepted
Rejected
Cancelled
session confirmed
Session Scheduled
cancel [any party]
session conducted
Session Completed


Skill Barter Platform
Register / Login
Manage Profile
Search Users by Skill
Send Barter Request
Accept / Reject Request
Message Matched User
Schedule Session
Rate and Review
View Barter History
Report User
Send Notifications
Manage Users
View All Sessions
View Platform Reports
Resolve Dispute
Registered User
Admin
Notification Service
<<include>>
<<include>>
<<include>>
<<include>>